Kaan Koc

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Education

University of California, Santa Cruz — **Expected Graduation: June 2026**
Bachelor of Science, Computer Science — GPA: 3.85

Relevant Coursework: Distributed Systems, Operating Systems, Data Structures, Algorithms, Web Applications

Experience

Software Engineering Intern (Backend) — **Sept. 2024 – Present**
Tech4Good Lab | Python, FastAPI, Node.js, PostgreSQL, LangChain, Docker — Santa Cruz, CA

- Delivered scalable backend systems supporting **AI content** platforms in production.
- Improved search relevance with a LangChain RAG pipeline on FastAPI + PostgreSQL, cutting latency by **~30%** through async I/O and caching.
- Designed and implemented an efficient API retry system in Node.js, **earning senior developer approval** and increasing response success rates to **over 90%.**
- **Built unit and integration tests** with **pytest/Jest,** increasing coverage by **~45%** and reducing production bugs, which cut incident reports by **~30%.**

Software Engineering Project Lead (Full-Stack) — **Sept. 2024 – June 2025**
Slug AI (Largest AI Club at UCSC) | Django, Express.js, React, MySQL, MongoDB, AWS — Santa Cruz, CA

- **Recruited and led 12+ developers**, training freshmen into active contributors; **6+ later secured SWE internships** after project experience.
- Directed **4 full-stack projects for local businesses**, delivering dashboards and AI analytics that cut manual workflows by **~30%**.
- Architected Django/Express.js backends with MySQL and MongoDB, and React/Next.js frontends, deployed into production environments actively used by business staff.
- Drove agile sprints, code reviews, and testing, raising project delivery speed by **~40% over previous semesters.**

Software Engineering Intern (Backend) — **June 2023 – Sept. 2023**
Papatia | C#, Java, Node.js, PostgreSQL, AWS, Docker — Remote

- Contributed to backend systems for e-commerce platforms serving **European clients**, powering customer transactions and product catalogs.
- Applied **object-oriented programming (OOP)** in C# and Java to build modular payment and order services.
- Collaborated through **GitHub pull requests**, merging **20+ successfully** reviewed contributions into production.
- Improved **API response times by ~25%** and deployed with Docker on AWS, ensuring **99.9% uptime**.

Awards & Certifications

Global Finalist – NASA Space Apps Challenge 2024, selected from 10,000+ participants worldwide — **Sept. 2024**

Projects

Interstellar Automated Visualizer | Python, Flask, D3.js, PostgreSQL, — **May 2025 – Aug. 2025**

- Built Flask backend with PostgreSQL pipelines for NASA datasets, cutting preprocessing time by **~45%.**
- Automated data ingestion from **NASA APIs**, eliminating manual handling.
- Designed interactive **D3.js dashboards** with dynamic filtering, boosting research analysis speed ~30% and earning **recognition from NASA Challenge juries.**

Shipwright AI | React, TypeScript, FastAPI, Node.js, MongoDB, GitLab CI/CD — **Mar. 2024 – June 2024**

- Built an **automation platform** that converts natural language prompts into production-ready software, reducing project setup time by **~80%**.
- **Automated GitLab repo creation, pipeline configs, and Dockerized deployments**, cutting time-to-deployment from **hours to minutes**.

Skills

Programming Languages: C/C++, Python, Java, C#, Kotlin, TypeScript, JavaScript, SQL, HTML/CSS, Assembly
Frameworks: FastAPI, Node.js, Express.js, Django, Next.js, React, React Native, Flask, Tailwind
Tools: REST APIs, pytest, Jest, LangChain, scikit-learn, Pandas, NumPy, Docker, Git, Figma, Linux
Databases/Cloud: MySQL, PostgreSQL, MongoDB, AWS, Google Cloud Platform